Name_____ Date_____ Period_____

LESSON 4-3 · SESSION 1 · GUIDED NOTES

Estimate with Benchmarks

Learning target – I can estimate a percent by comparing an amount to benchmark percents. · 6.AT.4

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
estimate *estimar*	A close answer that is not _____.	about 50% done
benchmark percent *porcentaje de referencia*	An _____ percent to use, like 10%, 25%, 50%, or 75%.	50% = one half
the whole *el entero*	The full amount. It counts as _____.	all 80 MB of a download

KEY IDEA

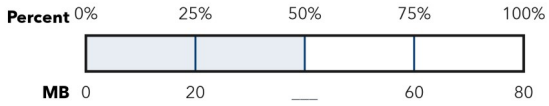
complete each sentence

- First name the _____. That amount is 100%.
- Find benchmark amounts (50% = _____, 25% = one fourth), then pick the one _____ to the part.

WORKED EXAMPLE

fill in the blanks as we solve

A game download is 80 MB. 41 MB are done. About what percent is done?



Step 1 The whole is ___ MB.

Step 2 50% of 80 MB is ___ MB.

Step 3 41 MB is just a little more than 40 MB.

Answer **About ____ of the download is done.**

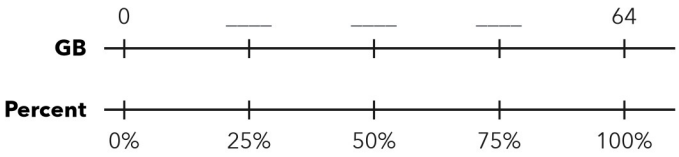
YOU TRY

use the example as a model

1. About what percent? Name the closest benchmark.

a. 12 out of 50 ≈ _____ % **b.** 59 out of 80 ≈ _____ % **c.** 8 out of 75 ≈ _____ %

2. A tablet holds 64 GB. 47 GB are used. Fill in the GB line, then estimate the percent used.



About_____ %

SUM IT UP

in your own words

To estimate a percent, I first name the ___ because ___.

_____

_____

Name_____ Date_____ Period_____

LESSON 4-3 · SESSION 2 · GUIDED NOTES

Estimate a Percent of a Number

Learning target – I can estimate a percent of a number using friendly percents and rounded amounts. · 6.AT.4

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
estimate *estimación*	An answer that is _____ to the exact amount.	about 60 bottles
compatible (friendly) numbers *números compatibles*	Numbers that are _____ to compute with in your head.	19% of 312 ≈ 20% of 300
ten percent *diez por ciento*	One tenth of the whole: divide by ___.	10% of 300 = 30

KEY IDEA

complete each sentence

- Round the percent and the whole to _____ (friendly) numbers.
- Find ____ first, then build: 20% is double, 5% is half of 10%.
- Write "about" because the answer is an _____.

WORKED EXAMPLE

fill in the blanks as we solve

Estimate 20% of 312 bottles.



Step 1 312 is about ____.

Step 2 10% of 300 is ___.

Step 3 20% is two 10% parts: 30 + 30 = 60.

Answer **About ___ bottles**

YOU TRY

use the example as a model

1. Estimate. Round first.

a. 21% of 198 ≈ _____ **b.** 52% of 60 ≈ _____ **c.** 11% of 703 ≈ _____

2. A bill is $38.40. Estimate a 15% tip.

Show your work.

About_____

SUM IT UP

in your own words

To estimate a percent of a number, I round to compatible (friendly) numbers because ___.

_____

_____

Name_____ Date_____ Period_____

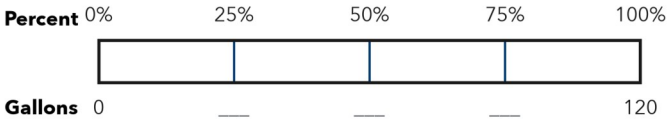
LESSON 4-3 · GUIDED NOTES · MODEL, REASON, EXPLAIN

Model It. Explain It.

SESSION 1 · ESTIMATE WITH BENCHMARKS

draw it · solve it · explain it

A water tank holds 120 gallons. It has 82 gallons in it. About what percent of the tank is full? Use the bar to find benchmark amounts.



a. Between which two benchmarks is 82 gallons?

b. Use 10% of 120 to give a closer estimate.

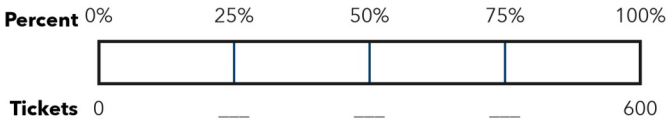
c. Explain how you know.

Challenge *Sam says 82 out of 120 is about 82%. Show why that is too big.*

SESSION 2 · ESTIMATE A PERCENT OF A NUMBER

draw it · solve it · explain it

A theater sold 588 tickets. About 26% were for kids. Estimate the number of kids' tickets two ways, using different compatible (friendly) numbers.



a. Estimate with 25% of 600.

b. Estimate with 25% of 560.

c. Which estimate is closer? Explain how you know.

Challenge *Sam says 150% of 58 is about 30, because 150% means take half. Show why that is too small.*

Check: Does my model match my numbers? Did I label units? Does my answer make sense?